Chapter 1. Vector Spaces

Linear combinations, independence, bases and dimension

Contents

1.1 Introduction **2**

1.2 Vector spaces **2**

1.3 Subspaces **4**

1.4 Linear combinations and systems of linear equations **5**

1.5 Linear dependence and linear independence **6**

1.6 Bases and dimension **7**

1.6.1 The Lagrange interpolation formula 11

1.7 Maximal linearly independent subsets * **11**

1.1 Introduction

Motivation. Eight rules obeyed by adding and scaling arrows are all that linear algebra uses.

Parallelogram law: with P at the origin, $x = (a_1, a_2)$, $y = (b_1, b_2)$, the diagonal is $x + y = (a_1 + b_1, a_2 + b_2)$; also $tx = (ta_1, ta_2)$.

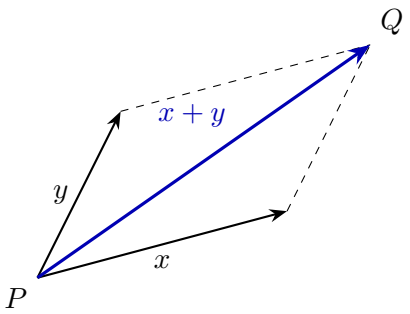


Figure. *Parallelogram law.*

For all vectors x, y, z and reals a, b:

1. $x + y = y + x$; 2. $(x + y) + z = x + (y + z)$; 3. there is 0 with $x + 0 = x$; 4. each x has y with $x + y = 0$;

5. $1x = x$; 6. $(ab)x = a(bx)$; 7. $a(x + y) = ax + ay$; 8. $(a + b)x = ax + bx$.

Lines and planes. Line through A, B (endpoints of u, v): $x = u + t(v - u)$. Plane through noncollinear A, B, C, with u, v from A to B, C: $x = A + su + tv$.

Example 1

$A = (-2, 0, 1)$, $B = (4, 5, 3)$: $x = (-2, 0, 1) + t(6, 5, 2)$.

Example 2

$A = (1, 0, 2)$, $B = (-3, -2, 4)$, $C = (1, 8, -5)$: $x = (1, 0, 2) + s(-4, -2, 2) + t(0, 8, -7)$.

1.2 Vector spaces

Motivation. A vector space is any set obeying the eight rules.

Definition (Vector space)

A **vector space** V over a field $\mathbb{F}$ (see Appendix C) is a set with an addition $V \times V \to V$ and a scalar multiplication $\mathbb{F} \times V \to V$ such that for all $x, y, z \in V$, $a, b \in \mathbb{F}$:
(VS 1) $x + y = y + x$; (VS 2) $(x + y) + z = x + (y + z)$;
(VS 3) there is $0 \in V$ with $x + 0 = x$ for all x; (VS 4) for each x there is y with $x + y = 0$;
(VS 5) $1x = x$; (VS 6) $(ab)x = a(bx)$;
(VS 7) $a(x + y) = ax + ay$; (VS 8) $(a + b)x = ax + bx$.
Elements of $\mathbb{F}$ are **scalars**, elements of V **vectors**.

An $m \times n$ **matrix** A has entries A_{ij} (row i, column j); A_{ii} are the **diagonal entries**, O is the **zero matrix**; A is **square** if $m = n$.

Example 1

$\mathbb{F}^n$, n-tuples with coordinatewise operations; in $\mathbb{C}^2$, $(1+i,2)+(2-3i,4i)=(3-2i,2+4i)$, $i(1+i,2)=(-1+i,2i)$.

Example 2

$\mathrm{M}_{m\times n}(\mathbb{F})$: $(A+B)_{ij}=A_{ij}+B_{ij}$, $(cA)_{ij}=cA_{ij}$.

Example 3

$\mathcal{F}(S,\mathbb{F})$ $(S\neq\emptyset)$: $(f+g)(s)=f(s)+g(s)$, $(cf)(s)=cf(s)$.

The **degree** of $f(x)=a_nx^n+\cdots+a_0$ is the largest k with $a_k\neq 0$; the **zero polynomial** has degree -1.

Example 4

$\mathsf{P}(\mathbb{F})$, polynomials with coefficientwise operations.

Example 5

Sequences in $\mathbb{F}$ with finitely many nonzero terms, termwise operations.

Example 6

On $\mathbb{R}^2$ with $(a_1,a_2)+(b_1,b_2)=(a_1+b_1,a_2-b_2)$, $c(a_1,a_2)=(ca_1,ca_2)$: (VS 1), (VS 2), (VS 8) fail.

Example 7

On $\mathbb{R}^2$ with $(a_1,a_2)+(b_1,b_2)=(a_1+b_1,0)$, $c(a_1,a_2)=(ca_1,0)$: (VS 3), (VS 4), (VS 5) fail.

Theorem 1.1 (Cancellation law for vector addition)

If $x+z=y+z$ in V, then $x=y$.

Proof. Take v with $z+v=0$ (VS 4). Then $x=x+0=x+(z+v)=(x+z)+v=(y+z)+v=y+(z+v)=y+0=y$ by (VS 2), (VS 3). ■

Corollary 1

The vector 0 of (VS 3) is unique.

Proof [Supplement]. If $0'$ also satisfies (VS 3), then $0+0'=0=0+0$ (by (VS 1), (VS 3)), so $0'=0$ by Theorem 1.1. ■

Corollary 2

The vector y of (VS 4) is unique.

Proof [Supplement]. If $x+y=0=x+y'$, then $y+x=y'+x$ by (VS 1), so $y=y'$ by Theorem 1.1. ■

0 is the **zero vector**, and the y of (VS 4) is the **additive inverse** $-x$.

Theorem 1.2

In any vector space V:
(a) $0x=0$ for each $x\in V$.
(b) $(-a)x=-(ax)=a(-x)$ for each $a\in\mathbb{F}$, $x\in V$.
(c) $a0=0$ for each $a\in\mathbb{F}$.

Proof.

(a) $0x+0x=(0+0)x=0x=0x+0=0+0x$, so $0x=0$ by Theorem 1.1.

(b) $ax+(-a)x=(a+(-a))x=0x=0$, so $(-a)x=-(ax)$ by Corollary 2. In particular $(-1)x=-x$, and $a(-x)=a((-1)x)=(a(-1))x=(-a)x$ by (VS 6).

(c) [Supplement] $a0+a0=a(0+0)=a0=a0+0$, so $a0=0$ by Theorem 1.1. ■

1.3 Subspaces

Motivation. Three closure checks decide whether a subset is a vector space in its own right.

Definition (Subspace)

A subset W of V is a **subspace** if it is a vector space over $\mathbb{F}$ with the operations of V. V and the **zero subspace** $\{0\}$ are subspaces.

(VS 1, 2, 5–8) are inherited by any subset.

Theorem 1.3

$W\subseteq V$ is a subspace iff (a) $0\in W$,
(b) $x+y\in W$ for $x,y\in W$, and (c) $cx\in W$ for $c\in\mathbb{F}$, $x\in W$.

Proof.

Step 1 (only if). (b), (c) are closure; the zero $0'$ of W has $x+0'=x=x+0$, so $0'=0\in W$ (Theorem 1.1).

Step 2 (if). Inverses: $-x=(-1)x\in W$ by (c) and Theorem 1.2; the other axioms are inherited. ■

The **transpose** has $(A^t)_{ij}=A_{ji}$; $(aA+bB)^t=aA^t+bB^t$. A is **symmetric** if $A^t=A$; symmetric matrices form a subspace of $\mathrm{M}_{n\times n}(\mathbb{F})$.

Example 1

$\mathsf{P}_n(\mathbb{F})$ (degree $\le n$) $\subseteq \mathsf{P}(\mathbb{F})$.

Example 2

$C(\mathbb{R})$ (continuous functions) $\subseteq \mathcal{F}(\mathbb{R},\mathbb{R})$.

Example 3

Diagonal matrices ($M_{ij} = 0$ for $i \neq j$) form a subspace.

Example 4

The **trace** $\operatorname{tr}(M) = M_{11} + \cdots + M_{nn}$ is linear; trace-zero matrices form a subspace.

Example 5

Matrices with nonnegative entries: not a subspace (not closed under $-1\cdot$).

Theorem 1.4

Any intersection of subspaces of V is a subspace of V.

Proof. Each member contains 0, $x + y$ and ax whenever it contains x, y; so does the intersection W. Apply Theorem 1.3. ■

A union of two subspaces is a subspace iff one contains the other. The **sum** $W_1 + W_2 = \{x + y : x \in W_1, y \in W_2\}$ is the smallest subspace containing both. V is the **direct sum** $W_1 \oplus W_2$ if $W_1 + W_2 = V$ and $W_1 \cap W_2 = \{0\}$.

1.4 Linear combinations and systems of linear equations

Motivation. Testing for a linear combination means solving a linear system.

Definition (Linear combination)

For nonempty $S \subseteq V$, v is a **linear combination** of vectors of S if $v = a_1u_1 + \cdots + a_nu_n$ for some finitely many $u_i \in S$ and scalars a_i (the **coefficients**). $0 = 0u$ is a linear combination of any nonempty S.

Example 1

Vitamin vectors of foods lie in $\mathbb{R}^5$; mixing foods combines them linearly.

Elimination. Swapping equations, scaling one by $c \neq 0$, and adding a multiple of one to another preserve solutions. Reduce until each leading coefficient is 1, its variable appears nowhere else, and leading variables move right; $0 = c \neq 0$ means no solution.

Example 2

$2x^3 - 2x^2 + 12x - 6 = a(x^3 - 2x^2 - 5x - 3) + b(3x^3 - 5x^2 - 4x - 9)$ means

$$a + 3b = 2, \quad -2a - 5b = -2, \quad -5a - 4b = 12, \quad -3a - 9b = -6,$$

solved by $a = -4$, $b = 2$. For $3x^3 - 2x^2 + 7x + 8$ the system reduces to $0 = 1$: not a combination.

Definition (Span)

The **span** $\operatorname{span}(S)$ of nonempty $S \subseteq V$ is the set of all linear combinations of vectors in S; $\operatorname{span}(\emptyset) = \{0\}$.

Theorem 1.5

$\operatorname{span}(S)$ is a subspace of V, and every subspace containing S contains $\operatorname{span}(S)$.

Proof. Clear for $S = \emptyset$. Otherwise $0 = 0z$, $\sum a_i u_i + \sum b_j v_j$ and $c\sum a_i u_i = \sum (ca_i)u_i$ are combinations (Theorem 1.3). A subspace $W \supseteq S$ is closed under the operations, so $W \ni \sum a_i u_i$. ■

Definition (Generating set)

S **generates** (**spans**) V if $\operatorname{span}(S) = V$.

Example 3

$(1,1,0), (1,0,1), (0,1,1)$ generate $\mathbb{R}^3$:

$$(a_1, a_2, a_3) = \tfrac{1}{2}(a_1 + a_2 - a_3)(1,1,0) + \tfrac{1}{2}(a_1 - a_2 + a_3)(1,0,1) + \tfrac{1}{2}(-a_1 + a_2 + a_3)(0,1,1).$$

Example 4

$x^2 + 3x - 2$, $2x^2 + 5x - 3$, $-x^2 - 4x + 4$ generate $\mathsf{P}_2(\mathbb{R})$ (solve a 3×3 system).

Example 5

The matrices $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ generate $\mathrm{M}_{2\times 2}(\mathbb{R})$. Matrices with equal diagonal entries cannot: their combinations keep that property.

1.5 Linear dependence and linear independence

Motivation. Independence means no vector is redundant: 0 has only the trivial representation.

Definition (Linear dependence)

$S \subseteq V$ is **linearly dependent** if there are finitely many distinct $u_1, \ldots, u_n \in S$ and scalars $a_1, \ldots, a_n$, not all zero, with $a_1 u_1 + \cdots + a_n u_n = 0$ (a **nontrivial representation** of 0). Any set containing 0 is dependent $(1 \cdot 0 = 0)$.

Example 1

Dependent in $\mathbb{R}^4$: $4(1,3,-4,2) - 3(2,2,-4,0) + 2(1,-3,2,-4) + 0(-1,0,1,0) = 0$.

Example 2

$M_1 = \begin{pmatrix} 1 & -3 & 2 \\ -4 & 0 & 5 \end{pmatrix}$, $M_2 = \begin{pmatrix} -3 & 7 & 4 \\ 6 & -2 & -7 \end{pmatrix}$, $M_3 = \begin{pmatrix} -2 & 3 & 11 \\ -1 & -3 & 2 \end{pmatrix}$: $5M_1 + 3M_2 - 2M_3 = O$.

Definition (Linear independence)

S is **linearly independent** if it is not linearly dependent: $\sum a_i u_i = 0$ with distinct $u_i \in S$ forces all $a_i = 0$.

$\emptyset$ is independent; $\{u\}$ is independent iff $u \neq 0$ (if $au = 0$, $a \neq 0$, then $u = a^{-1}(au) = 0$).

Example 3

$\{(1,0,0,-1), (0,1,0,-1), (0,0,1,-1), (0,0,0,1)\}$: $\sum a_i u_i = (a_1, a_2, a_3, a_4 - a_1 - a_2 - a_3) = 0$ forces all $a_i = 0$.

Example 4

$p_k = x^k + \cdots + x^n$ $(0 \leq k \leq n)$: in $\sum a_k p_k = 0$ the x^m-coefficient is $a_0 + \cdots + a_m$, so $a_0 = a_1 = \cdots = 0$.

Theorem 1.6

If $S_1 \subseteq S_2 \subseteq V$ and S_1 is linearly dependent, so is S_2.

Proof. A nontrivial representation of 0 by vectors of S_1 is one by vectors of S_2. ■

Corollary

If $S_1 \subseteq S_2 \subseteq V$ and S_2 is linearly independent, so is S_1.

Proof. Contrapositive of Theorem 1.6. ■

Theorem 1.7

Let S be linearly independent and $v \notin S$. Then $S \cup \{v\}$ is linearly dependent iff $v \in \text{span}(S)$.

Proof.

Step 1 (only if). In a nontrivial $\sum a_i u_i = 0$ ($u_i \in S \cup \{v\}$ distinct), independence of S forces $u_1 = v$, $a_1 \neq 0$; so $v = -a_1^{-1} \sum_{i \geq 2} a_i u_i \in \text{span}(S)$.

Step 2 (if). $v = \sum b_i v_i$ ($v_i \in S$) gives the nontrivial $\sum b_i v_i + (-1)v = 0$. ■

1.6 Bases and dimension

Motivation. Independent sets never outnumber generating sets, so all bases have one size: the dimension.

Definition (Basis)

A **basis** β for V is a linearly independent subset that generates V.

Example 1

$\emptyset$ is a basis for the zero space.

Example 2

$e_1 = (1, 0, \dots, 0), \dots, e_n = (0, \dots, 0, 1)$ form the **standard basis** of $\mathbb{F}^n$.

Example 3

The matrices E^{ij} (single 1 in position (i, j)) form a basis of $\mathrm{M}_{m\times n}(\mathbb{F})$.

Example 4

$\{1, x, \dots, x^n\}$ is the **standard basis** of $\mathsf{P}_n(\mathbb{F})$.

Example 5

$\{1, x, x^2, \dots\}$ is a basis of $\mathsf{P}(\mathbb{F})$.

Theorem 1.8

$\beta = \{u_1, \dots, u_n\}$ is a basis for V iff each $v \in V$ can be written uniquely as $v = a_1u_1 + \cdots + a_nu_n$.

Proof.

Step 1 (only if). Existence: β generates. $\sum a_iu_i = \sum b_iu_i \Rightarrow \sum(a_i - b_i)u_i = 0 \Rightarrow a_i = b_i$.

Step 2 (if) [Supplement]. β generates; uniqueness at $v = 0 = \sum 0u_i$ is independence. ■

Theorem 1.9

If V is generated by a finite set S, then some subset of S is a basis for V. Hence V has a finite basis.

Proof. If $S \subseteq \{0\}$, $\emptyset$ is a basis. Else grow $\{u_1\}$ $(u_1 \neq 0)$ by vectors of S keeping independence, until (finiteness) β has $\beta \cup \{v\}$ dependent for all $v \in S - \beta$; so $S \subseteq \mathrm{span}(\beta)$ (Theorem 1.7) and $V = \mathrm{span}(S) \subseteq \mathrm{span}(\beta)$ (Theorem 1.5). ■

Example 6

$S = \{(2, -3, 5), (8, -12, 20), (1, 0, -2), (0, 2, -1), (7, 2, 0)\}$: drop $(8, -12, 20) = 4(2, -3, 5)$ and $(7, 2, 0) = 2(2, -3, 5) + 3(1, 0, -2) + 4(0, 2, -1)$; basis $\{(2, -3, 5), (1, 0, -2), (0, 2, -1)\}$ of $\mathbb{R}^3$.

Theorem 1.10 (Replacement theorem)

Let V be generated by a set G of exactly n vectors and let L be linearly independent with exactly m vectors. Then $m \leq n$, and there is $H \subseteq G$ with exactly $n - m$ vectors such that $L \cup H$ generates V.

Proof. Induction on m; $m = 0$: $H = G$. For independent $L = \{v_1, \dots, v_{m+1}\}$, $\{v_1, \dots, v_m\}$ is independent (Corollary), so by induction $m \leq n$ and $\{v_1, \dots, v_m, u_1, \dots, u_{n-m}\}$ generates V for

some $u_i \in G$:

$$a_1 v_1 + \cdots + a_m v_m + b_1 u_1 + \cdots + b_{n-m} u_{n-m} = v_{m+1}. \tag{1.1}$$

If $n = m$ or all $b_i = 0$, $v_{m+1} \in \operatorname{span}\{v_1, \ldots, v_m\}$, contradicting Theorem 1.7. So $n \geq m + 1$ and, say, $b_1 \neq 0$. Solving (1.1) for u_1: $u_1 \in \operatorname{span}(L \cup H)$, $H = \{u_2, \ldots, u_{n-m}\}$. Then $\operatorname{span}(L \cup H) \ni v_1, \ldots, v_m, u_1, \ldots, u_{n-m}$, so it is V (Theorem 1.5). ■

Corollary 1

If V has a finite basis, every basis of V has the same number of vectors.

Proof. With $|\beta| = n$, every basis γ is finite with $m \leq n$ (Theorem 1.10 on finite subsets of γ); reversing roles, $n \leq m$. ■

Definition (Dimension)

V is **finite-dimensional** if it has a finite basis; the common number of vectors in a basis is its **dimension** $\dim(V)$. Otherwise V is **infinite-dimensional**.

Example 7

$\dim\{0\} = 0$.

Example 8

$\dim \mathbb{F}^n = n$.

Example 9

$\dim \mathrm{M}_{m\times n}(\mathbb{F}) = mn$.

Example 10

$\dim \mathrm{P}_n(\mathbb{F}) = n + 1$.

Example 11

$\mathbb{C}$ over $\mathbb{C}$ has dimension 1 (basis $\{1\}$).

Example 12

$\mathbb{C}$ over $\mathbb{R}$ has dimension 2 (basis $\{1, i\}$).

Corollary 2

Let $\dim(V) = n$.
(a) Every finite generating set has at least n vectors, and a generating set of exactly n vectors is a basis.
(b) Every linearly independent set of exactly n vectors is a basis.
(c) Every linearly independent subset of V extends to a basis.

Proof. Let β be a basis, $|\beta| = n$.

(a) Generating $G \supseteq$ basis H (Theorem 1.9), $|H| = n$ (Corollary 1); so $|G| \geq n$, and $|G| = n \Rightarrow G = H$.

(b) Theorem 1.10 with $G = \beta$, $|L| = n$: $H = \emptyset$, so L generates.

(c) For independent L, $|L| = m \leq n$, Theorem 1.10: $L \cup H$ generates with $|H| = n - m$, so $|L \cup H| \leq n$; by (a) it is a basis. ■

Example 13

By (a), the three polynomials of Example 4, Section 1.4, form a basis of $\mathsf{P}_2(\mathbb{R})$.

Example 14

By (a), the four matrices of Example 5, Section 1.4, form a basis of $\mathrm{M}_{2\times 2}(\mathbb{R})$.

Example 15

By (b), the four vectors of Example 3, Section 1.5, form a basis of $\mathbb{R}^4$.

Example 16

By (b), $\{p_0, \ldots, p_n\}$ of Example 4, Section 1.5, is a basis of $\mathsf{P}_n(\mathbb{F})$.

Theorem 1.11

If W is a subspace of a finite-dimensional V, then W is finite-dimensional and $\dim W \leq \dim V$; if $\dim W = \dim V$, then $W = V$.

Proof. Let $\dim V = n$, $W \neq \{0\}$. Grow an independent $\{x_1, \ldots, x_k\} \subseteq W$ from $x_1 \neq 0$; by Theorem 1.10 this stops at $k \leq n$, when no vector of W can be added. By Theorem 1.7 it generates W: $\dim W = k \leq n$. If $k = n$, it is a basis of V (Corollary 2(b)), so $W = V$. ■

Example 17

$W = \{(a_1, \ldots, a_5) \in \mathbb{F}^5 : a_1 + a_3 + a_5 = 0,\ a_2 = a_4\}$ has basis $\{(-1, 0, 1, 0, 0), (-1, 0, 0, 0, 1), (0, 1, 0, 1, 0)\}$, so $\dim W = 3$.

Example 18

The diagonal $n \times n$ matrices have basis $\{E^{11}, \ldots, E^{nn}\}$; dimension n.

Example 19

Symmetric $n \times n$ matrices: basis $\{A^{ij} : i \leq j\}$ (1 at (i, j), (j, i), else 0); dimension $\frac{1}{2}n(n+1)$.

Corollary

If W is a subspace of a finite-dimensional V, any basis for W extends to a basis for V.

Proof. A basis of W is an independent subset of V; apply Corollary 2(c) of Theorem 1.10. ■

Example 20

Even polynomials in $\mathsf{P}_{18}(\mathbb{F})$: basis $\{1, x^2, \ldots, x^{18}\}$, dimension 10.

Subspaces of $\mathbb{R}^3$: $\{0\}$, lines and planes through 0, $\mathbb{R}^3$ (dimensions 0–3).

1.6.1 The Lagrange interpolation formula

Let $c_0, \ldots, c_n$ be distinct scalars of an infinite field $\mathbb{F}$, and

$$f_i(x) = \prod_{k \neq i} \frac{x - c_k}{c_i - c_k} \qquad (0 \leq i \leq n),$$

the **Lagrange polynomials**: $f_i(c_j) = \delta_{ij}$. Evaluating $\sum a_i f_i = 0$ at c_j gives $a_j = 0$, so $\{f_0, \ldots, f_n\}$ is a basis of $\mathsf{P}_n(\mathbb{F})$ (Corollary 2(b)); evaluating $g = \sum b_i f_i$ gives $b_j = g(c_j)$:

$$g = \sum_{i=0}^{n} g(c_i) f_i \qquad (g \in \mathsf{P}_n(\mathbb{F})),$$

the **Lagrange interpolation formula**. So $\sum b_i f_i$ is the unique $g \in \mathsf{P}_n(\mathbb{F})$ with $g(c_j) = b_j$, and a nonzero $g \in \mathsf{P}_n(\mathbb{F})$ has at most n zeros.

1.7 Maximal linearly independent subsets *

Motivation. The Maximal Principle gives every space, even infinite-dimensional, a basis.

Definition (Maximal element; chain)

A member M of a family $\mathcal{F}$ of sets is **maximal** if no member of $\mathcal{F}$ other than M contains M. A collection $\mathcal{C}$ of sets is a **chain** if for all $A, B \in \mathcal{C}$, $A \subseteq B$ or $B \subseteq A$.

Example 1

In the power set of S, S is the (unique) maximal element.

Example 2

For disjoint nonempty S, T, $\mathcal{P}(S) \cup \mathcal{P}(T)$ has maximal elements S, T.

Example 3

Finite subsets of an infinite set have no maximal element.

Example 4

$A_n = \{1, \ldots, n\}$, $n \geq 1$, form a chain.

Maximal Principle (assumed; ≡ Axiom of Choice). If every chain $\mathcal{C} \subseteq \mathcal{F}$ has all its members contained in one member of $\mathcal{F}$, then $\mathcal{F}$ has a maximal member.

Definition (Maximal linearly independent subset)

$B \subseteq S$ is a **maximal linearly independent subset of** S if (a) B is independent and (b) the only independent subset of S containing B is B.

Example 5

By Example 2 of Section 1.4, any two elements of $S = \{2x^3 - 2x^2 + 12x - 6, x^3 - 2x^2 - 5x - 3, 3x^3 - 5x^2 - 4x - 9\}$ form a maximal linearly independent subset.

A basis of V is a maximal independent subset of V (Theorem 1.7).

Theorem 1.12

If S generates V and β is a maximal linearly independent subset of S, then β is a basis for V.

Proof. Need $S \subseteq \text{span}(\beta)$ (Theorem 1.5): $v \in S - \text{span}(\beta)$ would make $\beta \cup \{v\}$ independent (Theorem 1.7). ■

Theorem 1.13

If S is a linearly independent subset of V, some maximal linearly independent subset of V contains S.

Proof. Let $\mathcal{F}$ be the independent sets $\supseteq S$. For a chain $\mathcal{C} \subseteq \mathcal{F}$ put $U = \bigcup \mathcal{C}$ ($U = S$ if $\mathcal{C} = \emptyset$). If $\sum a_i u_i = 0$, $u_i \in A_i \in \mathcal{C}$ distinct, some $A_k \supseteq$ all A_i, so all $a_i = 0$: $U \in \mathcal{F}$. The Maximal Principle gives a maximal member of $\mathcal{F}$; it is maximal independent in V, as independent supersets contain S. ■

Corollary

Every vector space has a basis.

Proof. Apply Theorem 1.13 to $S = \emptyset$, and Theorem 1.12 with $S = V$. ■

All bases of a space have the same cardinality (cited).